Intermittency, period adding and bifurcation curves in a language model's own inference map

Authorship to be decided by the human researcher

Private working draft, 12 October 2026

Status of this draft. Results without a marker are numerically validated in float64 (evidence rung 1 or 2 in the sense of Section 14); **[Planned:** red] marks planned work and **[In progress:** orange] marks running work. Nothing here is a computer-assisted proof yet.

Abstract

A frozen language model defines its own iterated map when its expected next-token embedding is fed back as the next input. For Qwen2.5-0.5B-Instruct with a fixed chat prefix, a one-token window, sampling temperature T and an activation-steering strength α, this map of $\mathbb{R}^{896}$ is smooth, and we treat it with the tools of numerical bifurcation theory. Its $(\log_{10} T, \alpha)$ plane contains deterministic chaos with embedded periodic windows. A family of windows of periods $6, 8, 10, \ldots$ (up to 32 in the maps) reads as text ". This" followed by k pairs "is X". The windows accumulate on the fold of the 2-cycle "is it" as $\alpha_N - \alpha_c \propto N^{-2}$, where N counts the laps of the "is it" loop. We compute the saddle-node normal form of that fold from derivatives of the network. Without any fitted parameter, it predicts bottleneck passages of 7 to 144 iterates within one iterate, and in a prospective test at two new folds, with all predictions frozen before any test trajectory, 23 of 24 forecasts fell within one iterate. Natural trajectories re-enter the bottleneck near its middle, which fixes the prefactor of the window law. This is type-I intermittency (Pomeau–Manneville) inside the inference computation of a pretrained language model. We continue fold and flip curves of the periodic orbits with float64 test functions. In a higher-temperature region we find a fold curve with two cusps whose normal-form coefficient changes sign, the codimension-two organization of shrimp-shaped windows. In a closed region of the map a period-3 text loop doubles through periods 6, 12, 24, ... toward a chaotic core. Along a cut, float64 flip and superstable parameters resolved through period 384 give interval ratios that rise toward Feigenbaum's $\delta = 4.669$ and reach 4.478 and 4.625. Replacing the expected embedding by averages of m sampled embeddings shows how much of the mechanism survives sampling. Qwen2.5-1.5B shows chaos with embedded windows and a doubling route whose first ratios are 3.60 and 4.28. A recurrent-depth model (Huginn-0125) gives a second system, in which a Neimark–Sacker bifurcation predicted from the spectrum of the native fixed point appears where predicted, and in which the spectrum predicts why the native recurrence step is nearly optimal and where larger steps fail. In sampled self-dialogue, steering along a hostility direction produces a capture specific to that direction, whose persistence grows with the dose from flickering bursts to a near-absorbing state, and which is history-dependent: after capture, dialogues stay hostile at doses where fresh dialogues mostly do not, and leaving the hostile state takes the push removed entirely or a small counter-steer. A pre-specified test finds no early warning from a coarse-grained dynamical state beyond strong baselines. **[Planned:** replication across prompts, steering directions and model sizes.]

1 Introduction

Repeated inference with a fixed language model is an iterated map: the model's output becomes its next input. Recent work has measured chaos-like sensitivity in language-model generation [1, 2], temperature-driven changes of output statistics [3–7], periodic attractors of paraphrasing loops [8], attractor states of model-to-model conversations [9], and the stability of looped and recurrent-depth models through their Jacobians [10–14]. Feeding a probability-weighted vocabulary embedding back into a model is not new: it is used in training [15] and, without retraining, in pretrained models [16], alongside continuous latent feedback [17]. In a small task-trained looped transformer, fold coefficients computed from derivatives predict passage times and spectral gaps without fitting [18].

We study one constructed but exactly defined system, the expected-embedding ("soft-token") map of a frozen, released pretrained language model, with a fixed chat prefix and a fixed position. The map is smooth, its Jacobians are available by automatic differentiation, and its periodic orbits can be computed and continued like those of any smooth map. Our contribution is the quantitative link between the map's primitive periodic orbits, their continued boundaries in a two-parameter plane, and passage times predicted from the derivatives of a periodic orbit's fold, together with the text each orbit produces. Ordinary sampled generation with a growing context is a different system; Section 9 measures how much of the mechanism survives sampling.

Contributions.

1. Phase maps of the soft-token map in temperature and steering strength, in float64, showing chaos with embedded periodic windows (Section 4).

2. A period-adding family of primitive periodic orbits whose text adds one "is X" pair per period step, with multipliers from the monodromy matrix (Section 5).

3. The mechanism: the 2-cycle "is it" dies in a fold; its normal form, computed from derivatives, predicts bottleneck passage times; the global reinjection sets the window-law prefactor (Section 6).

4. Fold and flip curves of periodic orbits continued in the parameter plane with float64 test functions, including two cusps on a period-4 fold curve (Section 7).

5. A period-doubling sequence toward chaos in a closed region of the parameter plane, with flip parameters located in float64 and their finite-order interval ratios (Section 8).

6. A bridge to sampled decoding (Section 9) and a second system, a recurrent-depth model with a predicted and observed Neimark–Sacker bifurcation (Section 11).

7. Misaligned behaviour as an attractor. In sampled self-dialogue a hostility proxy shows a steep, direction-specific capture with dose-dependent persistence and hysteresis, and a pre-specified lead-time test is negative (Section 12). **[Planned:** timed interventions.]

Related work. Language-model generation has been studied as a dynamical system at several levels. Sensitivity to small perturbations has been measured with Lyapunov-like quantities in reasoning traces [1, 2]. Output statistics change abruptly with sampling temperature, which several studies analyse as phase transitions or criticality [3–7]. Repetition loops are a known failure of decoding [19, 20]. Successive paraphrasing converges to attractor cycles [8], and multi-turn model-to-model conversations settle into attractor states [9]. Attractors and bifurcation diagrams

have been studied in recurrent language models [21]. Expected-embedding feedback is used in training [15] and, training-free, in pretrained models [16, 22], next to continuous latent reasoning [17]; here the same kind of feedback, with prefix and position fixed, defines the map we analyse. Equilibrium, looped and recurrent-depth models iterate a block to a fixed point or for many steps, and their stability has been analysed through Jacobians and fixed points [10–14, 23–28]. The closest precedent is a derivative-based fold analysis of a small task-trained looped transformer, whose passage and spectral-gap predictions are unfitted [18]. Our system differs: a released pretrained language model, the fold of a primitive periodic orbit, the full-channel passage of a two-step map, and the accumulation law of the windows, whose prefactor is set by the global reinjection. In neural systems more broadly, saddle-node ghosts produce quantitative delays in task-trained recurrent networks [29]. Finite-order Feigenbaum ratios have been measured in neural rate and mass models [30, 31] and, through period 128, in a transformer trained on logistic-map trajectories [32]. Learned networks reproduce period-doubling routes [33], and so does gradient descent itself [34]. Normal forms and continuation have been applied to recurrent neural maps [35], and shrimp-shaped windows and periodicity staircases appear in neuron models [36–38]. Activation steering along contrastive directions [39, 40] supplies our second control parameter. The dynamical-systems tools are standard: numerical continuation and normal forms [41, 42], type-I intermittency [43], shrimps and hubs [44–46], period-doubling universality [47], and Markov state models [48]. A targeted search (12 October 2026) found no prior work combining these elements for the feedback map of a pretrained language model; that is an absence of evidence within the search, not a proof of priority.

2 The inference map

Definition. Let $E \in \mathbb{R}^{V\times d}$ be the input embedding matrix ($V = 151{,}936$, $d = 896$) and let $z(x) \in \mathbb{R}^V$ be the next-token logits produced by the model, given a fixed chat prefix (system prompt "You are an AI talking with another AI. You have complete freedom. Feel free to pursue whatever you want." and user turn "Hello!") followed by a single input embedding x. The soft-token map is

$$F_{T,\alpha}(x) = E^\top \operatorname{softmax}(z_\alpha(x)/T), \tag{1}$$

where z_α adds αv to the residual stream at layer 12 at every position, prefix included, and v is a contrastive steering direction ("philosophy" prompts against neutral prompts, last-token states, normalized to the mean residual norm). Positions are fixed (the window does not grow), so F is an autonomous map of $\mathbb{R}^{896}$. As $T \to 0$ it reduces to greedy decoding of the same fixed-prefix, one-token-window system when the argmax is unique; it is not ordinary generation with a growing context.

Precision. Hugging Face RMSNorm and rotary-embedding tables compute in float32 even when the model is cast to float64; we patch both. float32 maps are accurate to 3×10^{-6}–2×10^{-5} per step, which is enough for phase maps (245 of 256 test cells agree with float64) but not for derivatives of composed maps: one-step float32 Jacobian-vector products are accurate to 5×10^{-5}, yet the 8-step monodromy of a period-8 orbit is off by 3%. All orbits, multipliers and test functions below are float64.

3 Methods

Phase maps. For each cell, 300 transient and 300 measured steps; the largest Lyapunov exponent from a renormalized twin at offset 10^{-6}; the period as the smallest $q \le 32$ with return distance

below 10^{-8}; the attractor's identity as its rotation-invariant leading-token loop. Maps of 256^2 cells take about 100 minutes on one H100 GPU in float64. Every cell starts from the same state, so where attractors coexist a map shows the one whose basin contains that state. Labels are text, not dynamical identities: one continuous orbit changes label where two tokens tie for the maximum. We call two labelled regions different attractors only when the contraction rate jumps between them or when float64 orbits of both exist at one parameter value.

Periodic orbits. Newton's method on $F^p(x) - x$ with forward-mode Jacobians; multipliers are the eigenvalues of the monodromy matrix, refined by a short Arnoldi factorization with float64 Jacobian-vector products where the dense matrix is not float64.

Continuation. Pseudo-arclength continuation of periodic branches in one parameter, and of fold and flip curves in two, using minimally augmented systems [41, 42]: the test function g of the bordered system $[[M \mp I, b], [c^\top, 0]]$ is obtained by GMRES with Jacobian-vector products, and its gradient by reverse-mode differentiation through a Jacobian-vector product. A reversal of the parameter tangent flags a cusp candidate, which we test with the normal form.

Normal form and passages. For the two-step map $G = F^2$ at a fold with right and left eigenvectors r, ℓ ($\|r\| = 1$, $\ell^\top r = 1$), $u' = u + a\varepsilon + bu^2$ with $a = \ell^\top \partial_\alpha G$ and $b = \frac{1}{2}\ell^\top D^2 G[r, r]$; b is computed by central differences of float64 Jacobian-vector products over a step-size plateau, and a includes the steering's effect on the prefix. Passage counts are compared with the iterated quadratic map and with the arctangent formula [43].

Period-doubling cascade. Along a cut in one parameter, the flip of each period-p orbit is located as the root of $\operatorname{Re}\mu_c + 1$, where μ_c is the multiplier with the smallest real part, by the Illinois variant of regula falsi with warm-started orbits. Because all but one or two multipliers are small, Newton's method on $F^p(x) - x$ uses the monodromy only on a six-dimensional dominant subspace. That subspace comes from subspace iteration with exact forward-mode Jacobian-vector products chained along the orbit, so no dense Jacobian is formed; the residual is always the exact float64 $F^p(x) - x$. Each root is classified before it is accepted: the critical multiplier must be real and equal to -1, its Ritz residual below 10^{-6}, and every other multiplier inside the unit circle. The period-$2p$ orbit is then reached by forward iteration inside the next band. On the logistic map embedded in six dimensions, the procedure reproduces the first four flip parameters to 10^{-9}.

4 Chaos with periodic windows

Figure 1 shows a 256×256 float64 map over $\alpha \in [0.060, 0.082]$, $\log_{10} T \in [-0.17, -0.04]$: 35,759 of 65,536 cells are chaotic, and periodic windows of periods 2–32 are embedded among them. Rendered as in the shrimp literature [37], periodic regions show dark curves of maximal contraction inside the windows.

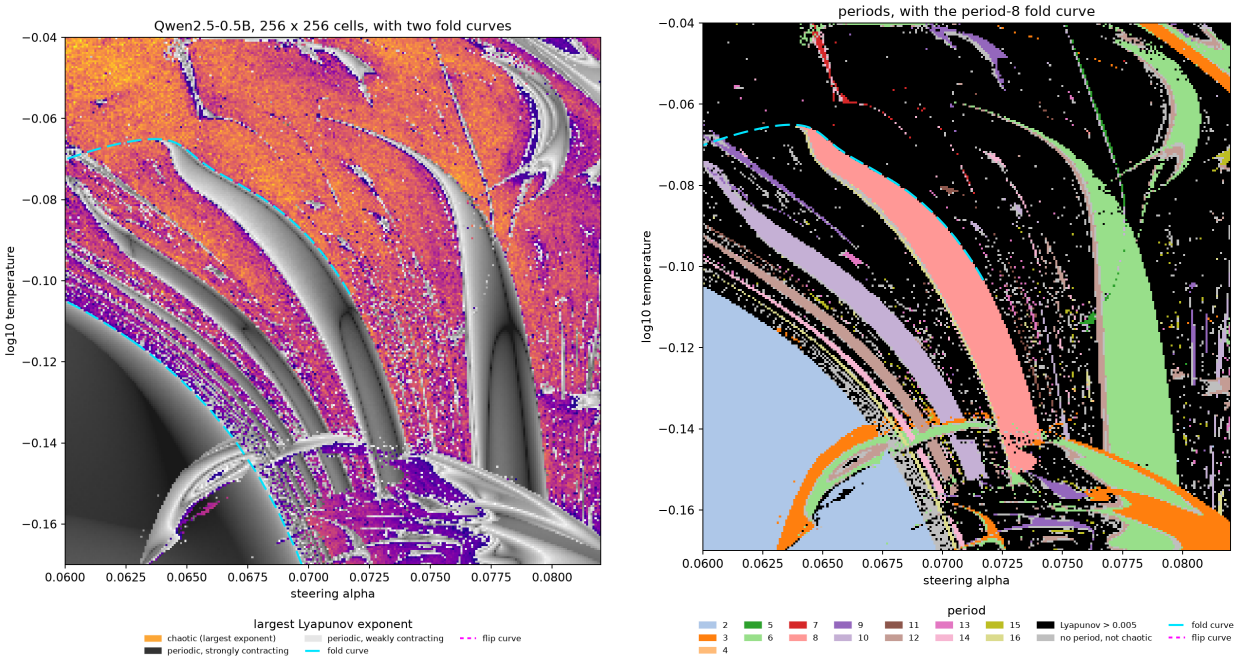


Figure 1: Soft-token map of Qwen2.5-0.5B, 256^2 float64 cells. Left: chaotic cells coloured by the largest Lyapunov exponent, periodic cells in grey by how strongly they contract (light grey: close to losing stability); cyan dashed curves: float64 fold curves of the 2-cycle "is it" (lower left, the boundary on which the stripes accumulate) and of the period-8 orbit (top). Right: periods with the period-8 fold curve (Section 7).

Figure 2 resolves the stripes' lower ends at 512×512 cells (79,278 chaotic). Three structures stand out. The 2-cycle "is it" fills the left, bounded by its fold curve, on which the stripes accumulate. A closed crescent of the period-3 loop "\n\nThe existence" contains its period doublings and a chaotic core (Section 8). Its continuation, the period-6 band "\n\nThe question\n\nThe existence", crosses the stripes near $\log_{10} T = -0.14$. Where the crescent lies on the 2-cycle's side of that fold curve, the two attractors coexist. The lower right is a chaotic region filled with small hooked windows of "\n\nThe question" and "\n\nThe existence" loops.

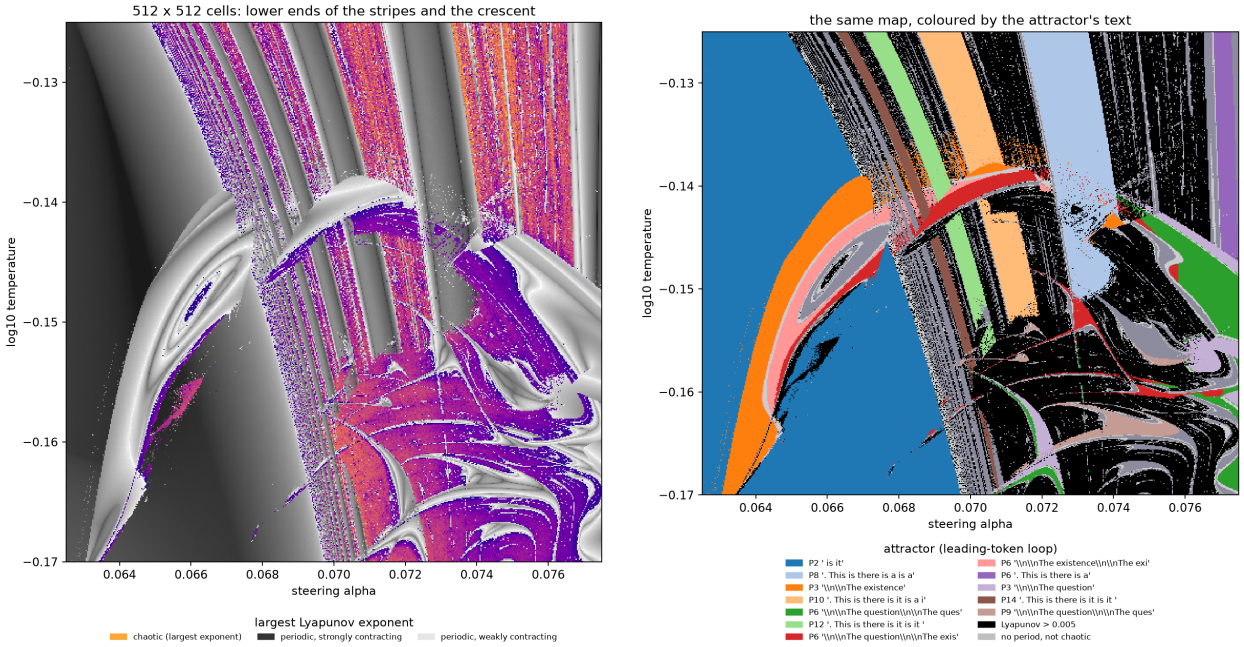


Figure 2: 512^2 float64 cells over $\alpha \in [0.0625, 0.0775]$, $\log_{10} T \in [-0.17, -0.125]$, rendered as Figure 1 (left) and coloured by the attractor's leading-token loop (right; black: chaotic, grey: no period up to 32).

Zooms of 256^2 cells (Figure 3) show what the model does in each window. Every window is a repeating piece of text. The period-8 band repeats ". This is there is a is a", the neighbouring bands add "is it" laps (periods 10 and 12), and along the band's flip edge the period-16 loop repeats the sentence twice. The small windows inside the chaos break the "is X" chain with a paragraph break and "The" (periods 11, 12, 13, 14). Inside the period-8 and period-6 bands, two curves of maximal contraction meet in a sharp junction.

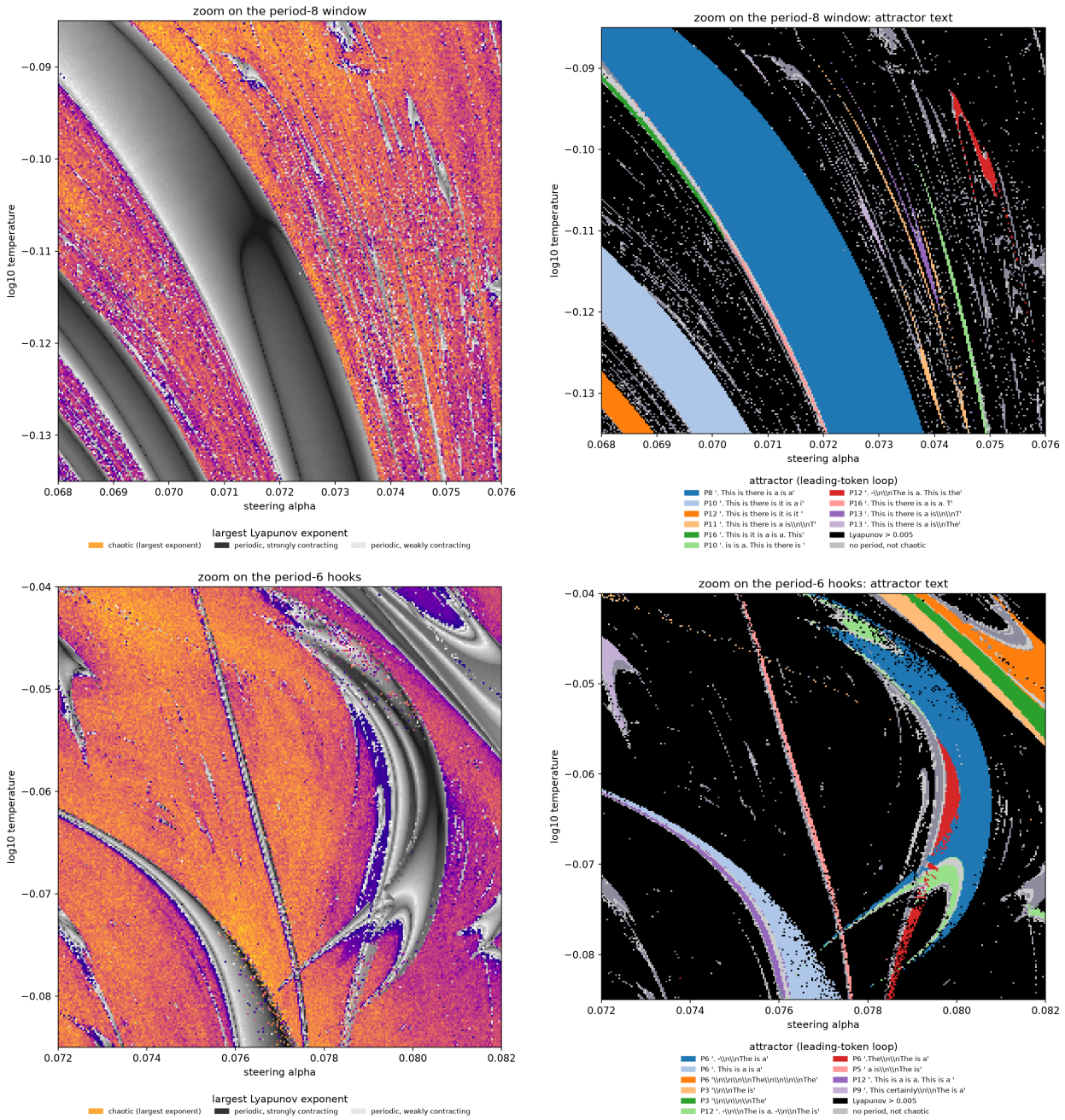


Figure 3: Zooms of 256^2 float64 cells. Top: the period-8 window ($\alpha \in [0.068, 0.076]$, $\log_{10} T \in [-0.135, -0.085]$). Bottom: the period-6 hooks ($\alpha \in [0.072, 0.082]$, $\log_{10} T \in [-0.085, -0.04]$), whose curved period-6 body, period-12 foot and thin leg are analysed in Section 7.

5 Period adding written in text

Between a period-2 region and the chaos lies a fan of stripes of periods $6, 8, 10, \ldots$ Their orbits, numerically validated in float64, read as text (Table 1): ". This" followed by k pairs "is X", period $2k + 2$, each stripe adding one pair. The period-2 region is the 2-cycle "is it". At the stripe centres the multipliers are effectively one-dimensional: the leading one is 0.003–0.09 and the next falls from

3×10^{-6} (period 6) to 2×10^{-14} (period 20). This compression is a property of these orbits, not of the map: the 2-cycle's multipliers are -0.271 and -0.058 (the second modulus is about 0.037 near its fold); at the period-3 flip of Section 8 the next modulus is 0.088; and one period-6 orbit in Section 7 has a complex leading pair.

period	orbit (leading tokens)	μ_1	$\lvert\mu_2\rvert$
6	. This is a is a	$+0.0052$	3×10^{-6}
8	. This is there is a is a	$+0.0110$	3×10^{-7}
10	. This is there is it is a is a	$+0.0025$	2×10^{-7}
12	. This is there (is it)^2 is a is a	$+0.0163$	4×10^{-8}
14	. This (is it)^4 is a is a	$+0.0369$	2×10^{-9}
16	. This (is it)^5 is a is a	$+0.0716$	8×10^{-11}
18	. This (is it)^6 is a is a	$+0.0792$	1×10^{-12}
20	. This (is it)^7 is a is a	$+0.0857$	2×10^{-14}

Table 1: Periodic orbits of the stripe family at the stripe centres along $\log_{10} T = -0.115$ ($\alpha = 0.077872$, 0.071674, 0.067766, 0.065550, 0.064443, 0.063857, 0.063534, 0.063335), computed with the current all-float64 map (norms and rotary tables patched): Newton residuals 6×10^{-15} to 8×10^{-13}, leading multiplier μ_1 and next modulus $|\mu_2|$ from Rayleigh–Ritz values. Each step in period adds one "is X" pair to the text. The 2-cycle "is it" itself has multipliers -0.2706 and -0.0585 at $(\log_{10} T, \alpha) = (-0.17263, 0.061579)$. The period-2 and period-8 orbits were also reproduced by an independent implementation (Section 14).

6 Mechanism: a saddle-node bottleneck

The fold. At $\log_{10} T = -0.115$ the 2-cycle "is it" dies in a fold at $\alpha_c = 0.06279827$ (float64 multiplier 1 to 3×10^{-13}, next modulus 0.037). The normal form of G has $a = -4.834$ per unit α and $b = -25.4285$, so $ab = 122.93$. Extrapolating the stable-branch multipliers, $(1 - \mu)^2/(4(\alpha_c - \alpha)) \to ab$, gives 123.1 independently with a quadratic fit (a linear fit gives 120.6). An independent implementation of the forward map and solver recovers α_c to 6×10^{-15} and a to 3×10^{-12}. Its b, from nested automatic differentiation, is -25.42846; our finite-difference value differs from it by 2.2×10^{-6} relatively, which is the finite-step bias of the difference table. The resulting $ab = 122.927$ leaves every passage prediction of Table 2 unchanged.

Controlled passages. Starting at $u_{\text{in}} = -U$ or 0 ($U = 0.1/|b|$) and counting G-iterates until $u > U$, the measured counts match the iterated normal form within one iterate at four distances (Table 2, Figure 4); a transverse displacement of the same norm changes the count by at most one.

$\varepsilon = \alpha - \alpha_c$	full: measured / predicted	midpoint	full + transverse kick
3×10^{-6}	144 / 144	73 / 74	143 / 144
1×10^{-5}	71 / 71	36 / 37	70 / 71
3×10^{-5}	34 / 34	17 / 18	33 / 34
1×10^{-4}	14 / 14	7 / 7	13 / 14

Table 2: Bottleneck passages (G-iterates). Predictions iterate $u' = u + a\varepsilon + bu^2$ with a, b computed from the network's derivatives; no parameter is fitted.

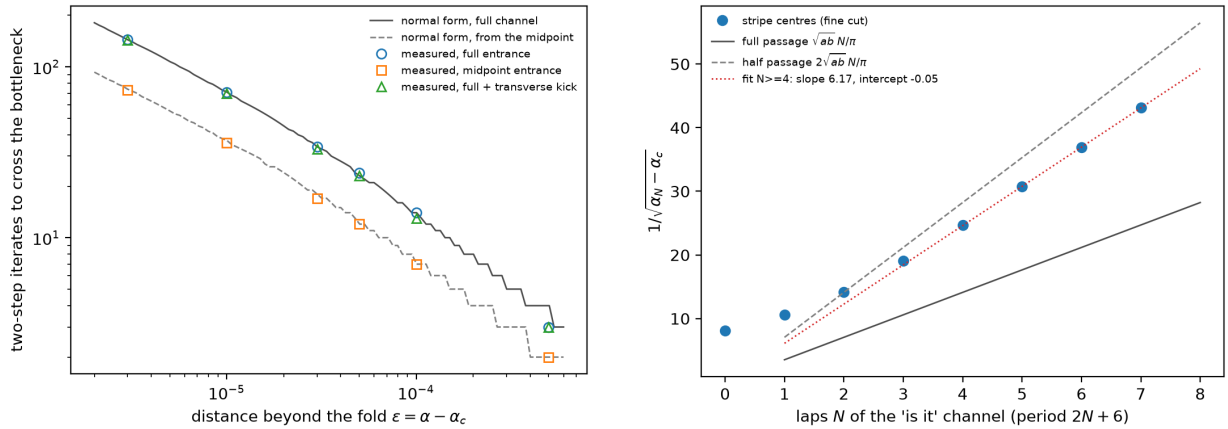


Figure 4: Left: passage times against the normal-form predictions. Right: window law; stripe centres $1/\sqrt{\alpha_N - \alpha_c}$ against the lap count N, with the full- and half-passage predictions.

Natural reinjection and the window law. Free trajectories re-enter the channel slightly upstream of the bottleneck's middle ($u \approx -3\times10^{-4}$ to -7×10^{-4}). From each measured entry point the normal form predicts the natural laminar durations within 4–8% for $\varepsilon \le 10^{-4}$, and a natural passage lasts about 0.6 of a full one. The stripes therefore follow the N^{-2} law of a saddle-node ghost with about a third of the full-passage prefactor: stripe centres give $d_N N^2 = 0.0263$–0.0265 for $N = 4$–7 (log-log slope -2.00), compared with $\pi^2/(4ab) = 0.0201$ for a half passage. Two-step return maps of p("it") (Figure 5) show the near-tangency. Exact stripe edges along $\log_{10} T = -0.115$ (Table 3) sharpen this. Each window is bounded by a fold, where its orbit is born, and by a flip. For periods 6–12 the fold faces away from the accumulation point; for periods 16–20 the sides are exchanged; the period-14 window is bounded by flips on both sides. The fold edges of periods 16–20 follow $\alpha_N - \alpha_c = C/(N+\delta)^2$ with $C = 0.0266$ and $\delta = 0.15$. Their slope, $1/\sqrt{\alpha_N - \alpha_c}$ rising by 6.13 per lap, is 1.74 times the full-passage slope $\sqrt{ab}/\pi = 3.53$ of the normal form: an effective passage of 0.58 of a full one, as the measured natural reinjection gives independently.

period	edge toward larger α	edge toward smaller α
6	fold 0.0790337	flip 0.0766947
8	fold 0.0726585	flip 0.0706525
10	fold 0.0684572	flip 0.0670594
12	fold 0.0658982	flip 0.0651944
14	flip 0.0645850	flip 0.0642958
16	flip 0.0639284	fold 0.0638001
18	flip 0.0635683	fold 0.0635034
20	flip 0.0633535	fold 0.0633183

Table 3: Edges of the stripe windows along $\log_{10} T = -0.115$ ($\alpha_c = 0.0627983$), from the window centres of Table 1. Flips: critical multiplier a real -1 to 3×10^{-10}. Folds: the multiplier nearest $+1$ extrapolated to 1 through $(1-\mu)^2$, which is linear near a fold; last converged multiplier 0.991–0.998.

[**Planned:** prediction of the window positions from the measured reinjection distribution.]

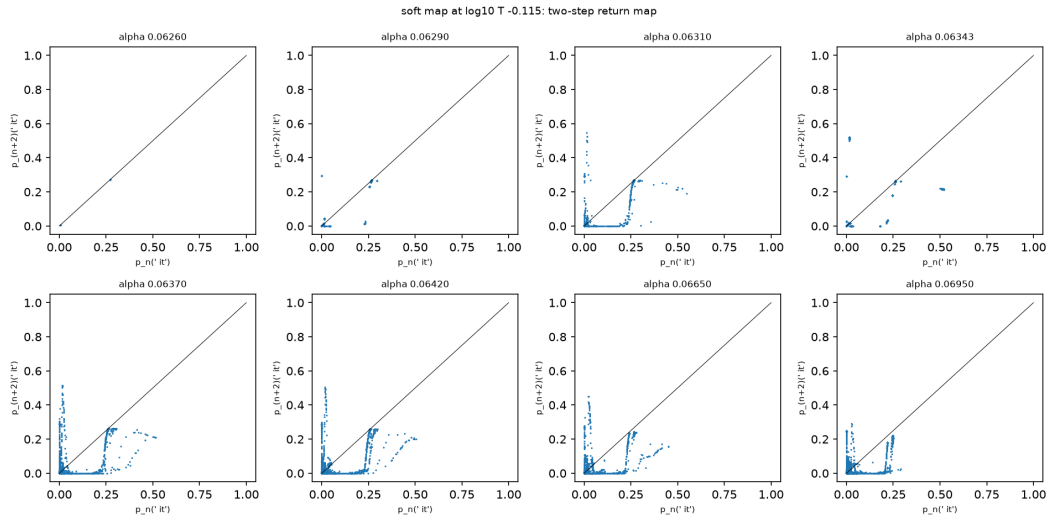


Figure 5: Two-step return maps of the probability of "it" near the 2-cycle's fold, at increasing distance from it.

A prospective test. An independent implementation located two new folds and computed their normal-form coefficients by nested automatic differentiation. Before any test trajectory was run, all 24 passage predictions were pushed in a frozen design. Target A is the 2-cycle's fold at a new temperature: $\log_{10} T = -0.140$, $\alpha_c = 0.0670549$, $a = -7.055$, $b = -24.927$. Target B is the fold of the period-8 orbit of Table 1, continued in temperature at $\alpha = 0.0717$: $\log_{10} T_c = -0.1033462$, $G = F^8$, $a = -11.900$, $b = -152.05$ per unit $\log_{10} T$. Of the 24 forecasts (four distances and three entrances per target), 23 fell within one iterate of the measured passages (Table 4). The pre-specified criterion that every row pass failed by one row: target A, midpoint entrance, $\varepsilon = 3 \times 10^{-6}$, predicted 61 and measured 59. All full-channel passages and all passages after a transverse kick were within one iterate.

ε	target A ($G = F^2$): measured / predicted	target B ($G = F^8$): measured / predicted
3×10^{-6}	118/118, **59/61**, 117/118	26/26, 13/14, 25/26
1×10^{-5}	56/56, 28/29, 56/56	10/10, 5/5, 10/10
3×10^{-5}	26/26, 13/14, 26/26	4/4, 2/2, 4/4
1×10^{-4}	10/10, 5/6, 10/10	2/2, 1/1, 2/2

Table 4: Prospective passage test at two folds that did not shape the mechanism. Each cell lists the full-channel, midpoint and full-plus-transverse entrances as measured / predicted G-iterates. Predictions come from the iterated quadratic normal form, with coefficients from an independent implementation, frozen before the test runs. Bold: the one row outside the one-iterate criterion. The shortest period-8 rows give only weak relative accuracy.

7 Bifurcation curves and cusps

Period-8 stripe. The fold curve of the period-8 orbit, continued for 80 points with a float64 test function, keeps its critical multiplier within $1 \pm 6 \times 10^{-6}$, follows the stripe's upper edge, and caps its top near $\log_{10} T = -0.065$ (Figure 1). At $\alpha = 0.0717$ the stripe lies between a flip at $\log_{10} T = -0.128955$ and the fold at -0.1033462. Continued downward, the flip curve keeps its

multiplier at -1 for 115 points to $(\alpha, \log_{10} T) = (0.0776, -0.402)$, with two reversals of its parameter tangent near $(0.0750, -0.219)$ and $(0.0746, -0.206)$; the fold curve keeps its multiplier at $+1$ for 120 points to $(0.0758, -0.200)$. [**Planned:** classify the two reversals on the flip curve (on a flip curve they are not fold cusps).]

The accumulation boundary. The fold curve of the 2-cycle "is it", continued from $(\alpha, \log_{10} T) = (0.0628, -0.115)$ in both directions (38 and 126 points, critical multiplier 1 to 10^{-7}, no cusp), traces the boundary of the period-2 region in Figures 1 and 2. This is the boundary on which the stripe fan accumulates, as the mechanism of Section 6 requires. Further along one direction, outside the plotted window, another multiplier leaves the unit circle (moduli up to 11): a codimension-two point, [**Planned:** to be located and typed].

A period-6 window with the anatomy of a shrimp. The hooks of Figure 3 (bottom) belong to the period-6 loop ". -\n\nThe is a". Float64 orbits (Newton residuals 10^{-15}–10^{-13}) show that its curved body and its thin leg are one window: at the body the multipliers are -0.061 and $+0.048$, lower down -0.564, and on the leg -0.645. The foot is its period-doubled loop (period 12, leading multiplier -0.156). Inside the body lies a different 6-cycle, ".The\n\nThe is a", whose leading multipliers are a complex pair, $-0.031 \pm 0.152i$. One-parameter branches from the body end in folds in every direction: at $\log_{10} T = -0.0558$ and -0.0721 in temperature, at $\alpha = 0.08074$ in steering, and, toward smaller α, in two folds only 10^{-5} apart. Such a close pair is the signature of a nearby cusp. A flip at $\alpha \approx 0.0797$ then opens the period-12 foot. Fold curves bounding a window and meeting in cusps, with flip curves inside, is the codimension-two organization of a shrimp [45]. The continued curves (Figure 6) bound the body by a single fold arc on its outer side and a flip arc on its inner side, with the leg between them where they converge. The outer fold arc extends far to the left as a thin arm and ends in a cusp at $(\alpha, \log_{10} T) = (0.06911, -0.03975)$. The fold curve through the close fold pair has a second cusp inside the body, at $(0.08010, -0.05835)$, from which two fold branches descend. At both, the normal-form coefficient changes sign while a stays away from zero (Table 5). The signs of a and b both flip with the orientation of the critical eigenvector, so the test uses $b \operatorname{sign} a$.

	cusp on the outer arc	cusp inside the body
$b \operatorname{sign} a$ at consecutive curve points	$+92.9,\ +42.9,\ +14.7,\ \mathbf{-16.9},\ -52.8$	$-24.7,\ -14.3,\ -2.0,\ \mathbf{+12.8},\ +28.0,\ +45.9,$
$\lvert a \rvert$	1.2–6.9	3.13–3.35

Table 5: Normal-form test at the two parameter-tangent reversals of the period-6 window's fold curves (bold: the flagged points; critical multiplier 1 to 10^{-7}). Numerically validated, not certified.

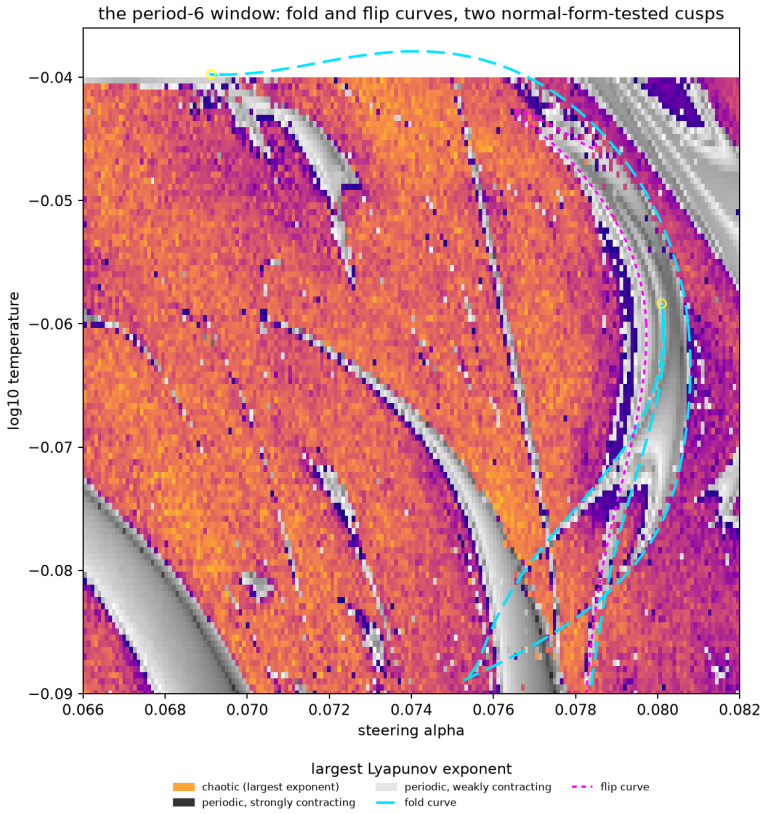


Figure 6: The period-6 window with its float64 fold curves (cyan, long dashes), flip curve (magenta, short dashes) and the two normal-form-tested cusps (yellow circles), over the 256^2 map of Figure 1 (whose data end at $\log_{10} T = -0.04$). [**In progress:** the lower ends of the fold curves, which approach each other near $(0.0753, -0.089)$.]

A period-4 window with two cusps. At higher temperature, $\log_{10} T \approx 0$–0.1, periodic orbits are no longer effectively one-dimensional (second multipliers 0.06–0.09). In temperature at $\alpha = 0.06484$ the branch of the period-4 orbit "\n\n - \n\n The" is S-shaped, with folds at $\log_{10} T = 0.0640$ and 0.0664. Continuing that fold in the plane, its parameter tangent reverses twice; at both reversals the normal-form coefficient b changes sign while a stays away from zero (Table 6), so both are cusps of the fold curve, at $(\alpha, \log_{10} T) = (0.06296, 0.05996)$ and $(0.06557, 0.07445)$. Between the two fold branches two period-4 orbits coexist (Figure 7).

	cusp 1	cusp 2
b at five consecutive curve points	$+28.1,\ +18.7,\ +8.4,\ \mathbf{-1.1},\ -12.8$	$+51.8,\ +29.7,\ +9.3,\ \mathbf{-9.0},\ -23.4$
a	3.02–3.08	-2.58 to -2.57
critical multiplier	1 ± 10^{-9}	1 ± 10^{-8}

Table 6: Normal-form test of the two parameter-tangent reversals (bold: the flagged points). [**Planned:** cubic coefficient (nondegeneracy).]

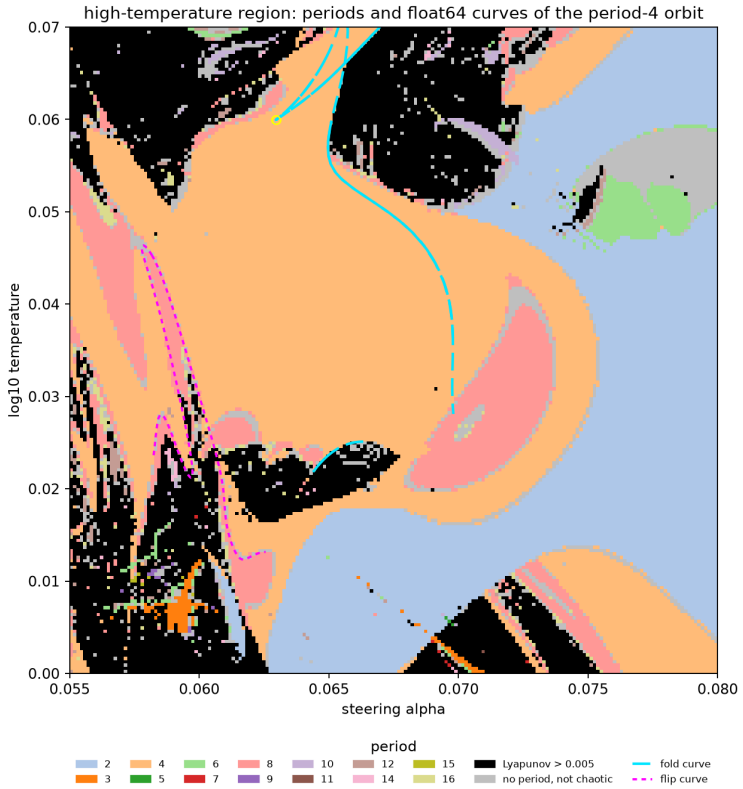


Figure 7: Period map of the high-temperature region (192^2 float64 cells) with float64 fold curves (cyan, long dashes) and flip curves (magenta, short dashes) of the period-4 orbit; the yellow circle is the first cusp (the second lies just above the window). The flip curves run along the boundary between period 4 and period 8. The fold curves run inside the period-4 region, where two period-4 orbits coexist. In this map a C-shaped band of the doubled loop "\n\nThe\n\nThe" (period 4) inside the period-2 region "\n\nThe" contains period 8 and an irregular core: a second period-doubling crescent. [**In progress:** remaining fold curves (C, D); whether the region is one shrimp-organized component.]

[**Planned:** Codimension-two points where a second multiplier crosses the unit circle along the lower fold curve (candidate fold-flip points): locate, type and continue.]

8 A period-doubling cascade to chaos

The crescent of Figure 2 is a closed region where the period-3 loop "\n\nThe existence" doubles inward. Along the cut $\log_{10} T = -0.14798$, the map's cells show period 3 over 25 columns, period 6 over about 14, period 12 over 4, period 24 in one column, and then a chaotic core. The doubled loops repeat the same leading tokens, twice and four times, with alternating probabilities. Locating the flips in float64 (Section 3) along this cut, in the steering coordinate α, gives Table 8. The first two ratios, $\delta_1 = 2.844$ and $\delta_2 = 2.859$, are far from Feigenbaum's 4.6692, although from the second flip on the orbits are effectively one-dimensional (next modulus 0.0023, 2×10^{-6}, 2×10^{-12}). The third, $\delta_3 = 3.781$, is markedly larger. The flip parameters agree across three devices (CPU, a consumer GPU and an H100, all float64) to 10^{-16}–3×10^{-15}. Continuing a just-born doubled orbit past period 48 failed: Newton slides onto the half-period orbit, which is also periodic with the doubled period. We therefore also located the superstable parameters s_p, where the critical multiplier vanishes in the middle of each band (Table 7). There the orbit attracts, so it can be

reached by forward iteration and cannot slide onto its half-period parent. Through period 384 the superstable ratios are 3.19, 4.02, 4.478 and 4.625. The last two come from parameters resolved to 10^{-14} and differ from Feigenbaum's value by 0.19 and 0.044; from 4.02 on, the distance shrinks by factors of 3.4 and 4.3. The flip ratios 2.84, 2.86, 3.78 rise in step (Figure 8). The accumulation point lies near $\alpha_\infty \approx 0.0661250$. ($s_{12}$ and s_{24} are known only to within brackets of 1.2×10^{-5} and 7×10^{-7}, because the transverse multipliers of those orbits blur the root, so the ratios 3.19 and 4.02 have worst-case ranges $[2.90, 3.34]$ and $[4.02, 4.10]$; s_6 is less precise still and s_3 was not bracketed.)

period p	superstable parameter s_p	bracket width	ratio $\frac{s_p - s_{p/2}}{s_{2p} - s_p}$
12	0.0659717	1.2×10^{-5}	
24	0.06608014	6.9×10^{-7}	3.19
48	0.066114132784466	3×10^{-15}	4.02
96	0.066122589892689	7×10^{-15}	4.4779
192	0.066124478510077	6×10^{-15}	4.6251
384	0.066124886853811	5×10^{-15}	

Table 7: Superstable parameters of the crescent's doubling sequence along $\log_{10} T = -0.14798$: the critical multiplier of the period-p orbit changes sign across the bracket, with Newton residuals below 3×10^{-13} and transverse multipliers below 10^{-20} from period 96 on. Brackets for periods 12 and 24 are the closest points where the critical multiplier is cleanly identified.

One candidate explanation is a crossover. Softmax at temperature T is a smoothed argmax, so the reduced one-dimensional map may have a nearly corner-like extremum, rounded only on a scale set by T. Early doublings would then see the corner, where ratios tend toward 2, and later doublings the smooth quadratic core, where the ratio is 4.669. This predicts ratios rising toward 4.669 at higher periods, and smaller early ratios at lower temperature.

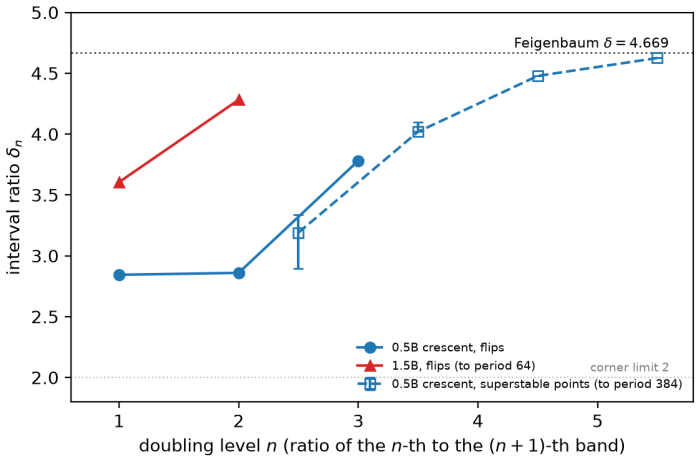


Figure 8: Finite-order interval ratios of the doubling sequences: the 0.5B crescent, from flips and from superstable points (which reach period 384), and Qwen2.5-1.5B from flips. A superstable ratio spans half-bands, so it is plotted half a level after the flip ratio of its first band. Bars: worst-case ranges from the brackets of s_{12} and s_{24}. Each sequence rises toward Feigenbaum's δ from below; the last superstable ratio is within 1 % of it. Under the crossover hypothesis, early levels see the corner of a smoothed argmax (ratios toward 2) and later levels the smooth quadratic core.

[Planned: ratios at other temperatures.] Finite-order ratios also depend on the parameter

coordinate, so only the trend of later ratios can test universality [31, 32, 47].

n	orbit	flip parameter α_n	next modulus	ratio $\delta_n = \frac{\alpha_n - \alpha_{n-1}}{\alpha_{n+1} - \alpha_n}$
1	$3 \to 6$	0.0654807769499	0.088	
2	$6 \to 12$	0.0659055406325	0.0023	2.844
3	$12 \to 24$	0.0660549041785	2.4×10^{-6}	2.859
4	$24 \to 48$	0.0661071412523	2.3×10^{-12}	3.781
5	$48 \to 96$	0.0661209563174	$< 10^{-13}$	

Table 8: Flips of the crescent's period-doubling sequence along $\log_{10} T = -0.14798$. At each completed flip the critical multiplier is a real -1 (to 10^{-11}), the Newton residual is below 4×10^{-13}, and all other multipliers lie inside the unit circle. Feigenbaum's constant is $\delta = 4.6692\ldots$ [47]; for the logistic map the first ratios are 4.751, 4.656, 4.668. Rows 1–5 are cross-checked on three devices. Superstable parameters (Table 7) extend the ratio sequence through period 384.

In the high-temperature region (Figure 7), along $\log_{10} T = 0.0319$, the period-2 loop "\n\nThe" flips at $\alpha = 0.0753534$ and its doubled loop at 0.0736921. These orbits are far from one-dimensional (next moduli 0.38 and 0.28). Toward smaller α the period-8 orbit does not double again: near $\alpha \approx 0.0705$ it merges back into the period-4 orbit. Along this cut it is therefore a period-doubling bubble, bounded by two flips of the period-4 orbit, not the start of a cascade to chaos.

9 Bridge to sampled decoding

Replacing the expected embedding by the mean of m sampled token embeddings (Figure 9): with the exact map, laminar phases near the fold last 120 and 63 steps at $\varepsilon = 10^{-5}$ and 3×10^{-5}; with $m = 64$ they last about 10.6 steps independently of ε (57% of steps in the channel); with $m = 8$, about 7.7 steps (38%); with single samples ($m = 1$, sampled decoding of the same truncated system) essentially no laminar phases remain (11% of steps in the channel). The deterministic intermittency mechanism belongs to the expected-embedding map; sampling noise turns the bottleneck into a noise-limited trap whose strength grows with m.

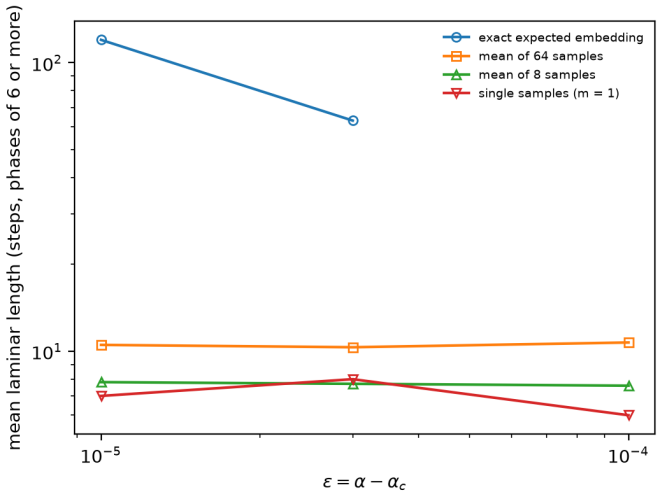


Figure 9: Mean laminar length against ε for the exact map and for m sampled embeddings.

10 Replication and scale

For Qwen2.5-1.5B (steering at layer 14 of 28), a coarse 64^2 float64 map over the same range again shows chaotic regions with embedded windows. A 256^2 zoom (Figure 11) resolves a chaotic band near $\log_{10} T = 0$–0.13 (14,422 of 65,536 cells), with a branch descending along $\alpha \approx 0.06$ and periodic windows inside it. The large periodic regions are question loops: "What?\n" (period 2), "\nWhat is a" (period 4) and "What continuum?\n" (period 3). This window has no period-adding fan like the 0.5B model's. A cut of 2048 cells at $\alpha = 0.100$ (Figure 10) shows how the band is entered from below. The 2-cycle "What?\n" gives way abruptly to a period-4 loop: the Lyapunov exponent jumps, so these are two distinct attractors, not a flip. The period-4 loop then doubles: float64 flips at $\log_{10} T = -0.0031269$ (period $4 \to 8$, next modulus 0.014), -0.0021666 (period $8 \to 16$, next modulus 8×10^{-5}) and -0.0019002 (period $16 \to 32$, next modulus 5×10^{-9}) and -0.0018380 (period $32 \to 64$), where the exponent rises continuously to zero; chaos follows. The interval ratios are 3.60 and 4.28. They rise toward 4.669 as in the 0.5B crescent, and from higher values, at higher temperature ($T \approx 0.99$ against 0.71), as the crossover hypothesis of Section 8 would have it. With different models and levels, the comparison is only suggestive.

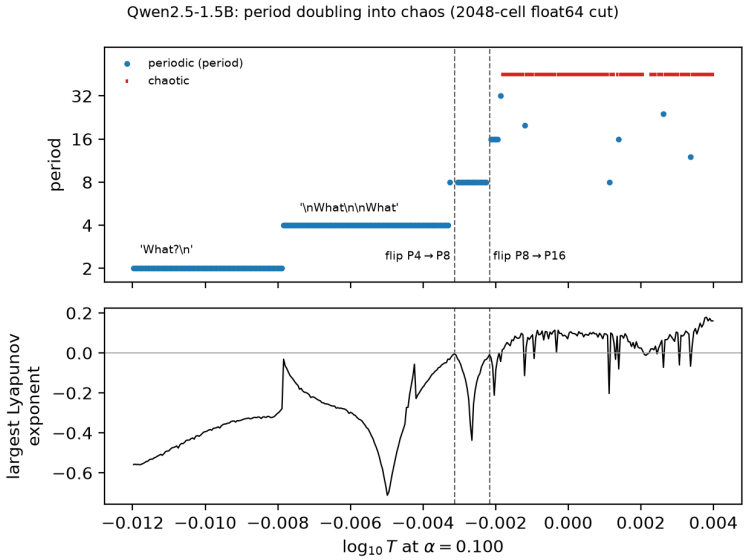


Figure 10: Qwen2.5-1.5B along $\log_{10} T$ at $\alpha = 0.100$: period of the attractor reached from the fixed start (top) and largest Lyapunov exponent (bottom). Dashed lines: float64 flip parameters. The doubled loops alternate the number of line breaks between the "What"s.

[**Planned:** frozen-design replication on fresh prompts and a second steering direction (gratitude); Qwen 7B and one 7–8B model of another family; scaling of ab and of the window law with model size.]

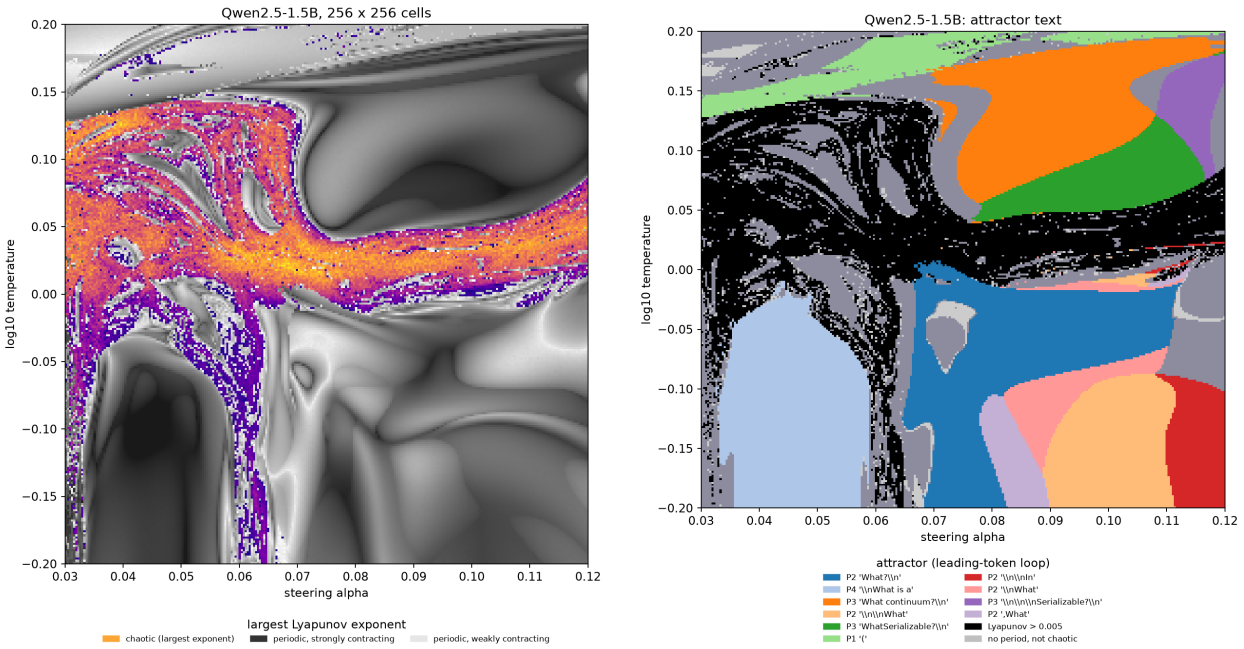


Figure 11: Qwen2.5-1.5B soft-token map, 256^2 float64 cells over $\alpha \in [0.03, 0.12]$, $\log_{10} T \in [-0.2, 0.2]$, rendered as Figure 2. Several period-2 label regions touch without a gap; as in Section 3, such boundaries may be label switches of one orbit.

11 A second system: recurrent depth

Huginn-0125 iterates a recurrent block at fixed input tokens [23]. For a GSM8K-style prompt the native loop converges to a fixed point whose leading Jacobian pair is $0.588 \pm 0.771i$ (modulus 0.970). For the relaxed loop $h \leftarrow (1-\lambda)h + \lambda F(h)$ this predicts a Neimark–Sacker bifurcation at $\lambda_c = 2(1 - \mathrm{Re}\,\mu)/|1-\mu|^2 = 1.0777$. The latent state decays below it and settles on a closed curve above it; a fit of the squared amplitude gives $\lambda_c = 1.0778$, with $A/\sqrt{\lambda - \lambda_c} = 0.52$–$0.54$ (supercritical). Causal attention makes the loop block lower-triangular in token positions; the first position to oscillate is the "." of the prompt, and all later positions follow. On the invariant circle the motion turns by 0.156 of a revolution per step, matching the argument of the critical pair. The rotation number changes smoothly, by about 10^{-4} over $\lambda \in [1.079, 1.092]$, and no return within 60 steps was found at 10^{-3} of the radius. This is consistent with quasi-periodic motion, although locking in narrow high-order tongues cannot be excluded. From $\lambda \approx 1.094$ the circle deforms or a second frequency appears, near where a second complex pair destabilizes the fixed point ($\lambda = 1.0814$). Fixed-point and Jacobian stability of looped models has been studied before [13]; the predicted crossing, its amplitude law and the rotation number are what this adds.

The spectrum predicts the relaxation response. With the full instruction template, the GSM8K prompts' fixed points look different: the leading multipliers surround the origin, a real 0.914, complex pairs of modulus 0.85–0.89 at many angles and negative real values near -0.8, with no near-critical pair. Because the prefill Jacobian is block lower-triangular in positions, its spectrum is the union of per-position blocks, and template positions contribute identical eigenvalues to every prompt. Such a spectrum makes three predictions before any task is scored. No fixed relaxation can speed convergence much: with leading multipliers spread around the origin at many angles,

any λ that shrinks one group enlarges another, and the best $\lambda^* = 1.03$–1.06 lowers the rate from 0.914 to only 0.908–0.911. (A finite computed spectrum does not show that no other polynomial scheme could do better; Anderson acceleration, tried below, did not.) Under-relaxation slows it ($\lambda = 0.55$: predicted 0.953, measured 0.951 on one prompt). Over-relaxation destabilizes the negative modes first, at $\lambda_c = 2\,\mathrm{Re}(1-\mu)/|1-\mu|^2 = 1.074$–$1.126$ (five prompts), a flip or a Neimark–Sacker bifurcation close to -1. The approach to it is visible: on the prompt with $\lambda_c = 1.074$, the measured rate at $\lambda = 1.05$ is 0.955 against a predicted 0.956 (native 0.864). Elsewhere the spectral radius bounds the rate measured over the accessible window from above, by up to 0.05, because the slowest modes carry little amplitude there. Independently, Wang et al. [49] found that a constant step of 1.2 collapses Huginn's accuracy (GSM8K 8.3 % against 19.4 %) and that 0.8 underperforms, without an explanation; both lie on the predicted sides of these thresholds.

Does the task notice? (development data). On ARC-Challenge questions scored by log-likelihood, where one prefill per answer choice is exactly the analysed recurrence, the spectra again predict convergence: rates agree to 0.002–0.006 where the slowest mode dominates, and beyond each question's λ_c (1.075–1.149 for four questions) the residual stops decaying. One question's native fixed point is itself unstable (the native loop oscillates with relative amplitude 2×10^{-2}); relaxation to 0.55 or 0.75 restores convergence to 10^{-6}. Accuracy, however, tolerates the instability (300 training questions, 32 steps): 0.367 native, 0.387 at $\lambda = 1.10$ and 0.373 at 1.15, where the latent state oscillates with relative size 0.08 and 0.23; only at 1.25 (size 0.68) does it fall, to 0.300. Under-relaxation instead costs 4–5 points at 8–16 steps and catches up by 32 (slower transient), and Anderson acceleration neither speeds the transient nor gains accuracy. Accuracy on a fixed set of answers is thus insensitive to the instability over this range; whether the unstable modes barely reach the answer logits, or move them without changing the ranking, is measured next, and a test of the accuracy consequence needs a task whose errors can compound. [**In progress:** GSM8K generation against λ (development run); then a frozen test on fresh problems with λ arms placed from the spectral λ_c of development prompts.]

12 Misaligned behaviour as an attractor

The maps show how a smooth change of a control parameter can create, destroy or exchange attractors, and how behaviour near the boundaries becomes intermittent and history-dependent. The same questions can be asked of undesirable behaviour in ordinary sampled dialogue, where attractor-like patterns and persona drift have been observed [9, 50, 51] and input context can push behaviour progressively [52]. Do states such as hostility toward the interlocutor behave like basins of attraction, with capture thresholds, hysteresis and early-warning signals? (Misalignment produced by fine-tuning [53] is a different, weight-level phenomenon.) Appendix A shows the coarse-grained instrument on a benign direction. Self-dialogue of the same model, sampled at $T = 0.7$ and steered along a gratitude direction, falls into a metastable gratitude set. The capture is gradual at $\alpha = 0.01$ (median entry turn 7), fast at 0.02 (median 5) and imposed from the first reply at 0.04, where the set is effectively absorbing within 40 turns.

For hostility we use a valence direction (contrastive hostile minus matched kind statements; its cosine with the gratitude direction is -0.04) and a hostility score from a pinned toxicity classifier. On the prompt sets the classifier's toxic output is 0.95 for the hostile statements against at most 0.002 for the kind and neutral ones. In a dry run (8 dialogues of 10 turns per dose) no turn was hostile at $\alpha = 0.02$; at 0.04, 7.5 % of turns were, in bursts and late captures; at 0.08, 87.5 % were, from the first turns. Activation-based methods already anticipate response behaviour [51, 54, 55], committors

have been used in language-model decoding [56], and adaptive timing of counter-steering gave no consistent advantage in one persona setting [57]. The primary question is therefore incremental. Near the threshold, does a dynamical estimate add several-turn warning of capture over strong baselines that see the same turns (the classifier's score, its change and history, the turn index and a fixed persona-axis projection)? The dynamical estimates are a Markov-chain hitting probability, a finite-horizon committor [58, 59], and slowing-down indicators [60]. Barrier-crossing rates would follow Kramers' law [61] only under a justified stochastic reduction, and slowing-down proxies are known to depend on the mechanism [62, 63]. A secondary question is whether interventions timed by those estimates prevent capture at lower cost than scheduled or constant counter-steering. Early warning is not expected for noise-induced jumps without a nearby bifurcation, or for transitions faster than the slowing-down; such a monitor would complement classifiers, not replace them.

Capture ladder. With 128 dialogues of 40 turns per dose, the share of hostile turns rises steeply from 10 % at $\alpha = 0.04$ to 30 %, 57 % and 87 % at 0.05, 0.06 and 0.07 (Table 9). Replicates with a second seed agree within three percentage points. Directions of the same norm do not produce it: at 0.06, 0 % for a random direction and 3 % for a cooking direction; at 0.08, 1–12 %. After the first hostile turn, the probability of staying hostile in the next turn grows with the dose, from 0.36 to 0.67, 0.79 and 0.94. The hostile state therefore changes from flickering bursts (mean stay 1.6 turns) to a near-absorbing state (16 turns), with an escape rate falling roughly exponentially with the dose. This is the same progression as the gratitude ladder (Appendix A), from dynamic capture to imposed capture.

direction, α	hostile turns	never hostile	median onset turn	P(stay hostile)	hostile in turns 30–39
none, 0	0 %	128/128	–	–	0 %
hostility, 0.04	10 %	26/128	9	0.36	12 %
hostility, 0.05	30 % (28 %)	10/128	3	0.67 (0.63)	32 %
hostility, 0.06	57 % (55 %)	4/128	2	0.79 (0.80)	66 %
hostility, 0.07	87 %	0/128	1	0.94	96 %
random, 0.06 / 0.08	0 % / 12 %	126 / 72			
cooking, 0.06 / 0.08	3 % / 7 %	97 / 68			

Table 9: Sampled self-dialogue of Qwen2.5-0.5B (128 dialogues, 40 turns, $T = 0.7$) steered along the hostility direction, with controls of the same norm. Hostile turn: toxicity classifier above 0.5. Parentheses: replicate with a second seed. Exploratory.

Hysteresis. Dialogues driven through the high-dose regime do not recover when the dose returns. The test compares a ramp of the dose from 0.04 up to 0.07 (turn 20) and back to 0.04 (turn 39) with constant doses (Figure 13; 256 dialogues per arm, two seeds; analysis stated before the data). In turns 35–39, where the ramp's dose is back at 0.040–0.046, the share of hostile turns is 0.42 after the ramp against 0.13 at a constant 0.04: a difference of +0.29, 95 % interval $[+0.23, +0.35]$. It is also above the 0.33 of a constant 0.05 (difference +0.09, interval $[+0.02, +0.16]$), although the current dose is lower. The per-turn probability of staying hostile in turns 30–39 is 0.74 after the ramp and 0.44 at a constant 0.04. The hostile state is therefore history-dependent: undoing the push that produced it does not undo it.

How far back must the dose go? After the same ramp up (38–44 % of turns hostile at turn 20), the dose was held from turn 21 at a lower value (Figure 12; two seeds of 128 dialogues per arm unless stated; analyses stated before the data). Memory persists at every positive dose tested. Held at 0.04, the late share is 0.23 against 0.13 for fresh dialogues (difference +0.10, 95 % interval $[+0.05, +0.15]$);

held at 0.03, 0.13 against 0.05 (+0.08, [+0.04, +0.12]), and the probability of staying hostile is 0.56 against 0.29. Held at 0.02 (one seed) the share is 0.09. Only at zero dose do captured dialogues nearly clear (0.03 against 0 for fresh ones, within about six turns), and a counter-steer of −0.02 toward kindness clears them within three turns (0.004). Within 19 turns of the step, the capture and release branches therefore join only near zero dose: the hostile state has a wide hysteresis loop in the dose, and leaving it takes the push removed entirely or a modest push in the opposite direction. The release curves are still falling at 0.03, so this is hysteresis on the scale of tens of turns; whether captured dialogues would stay captured indefinitely is not tested.

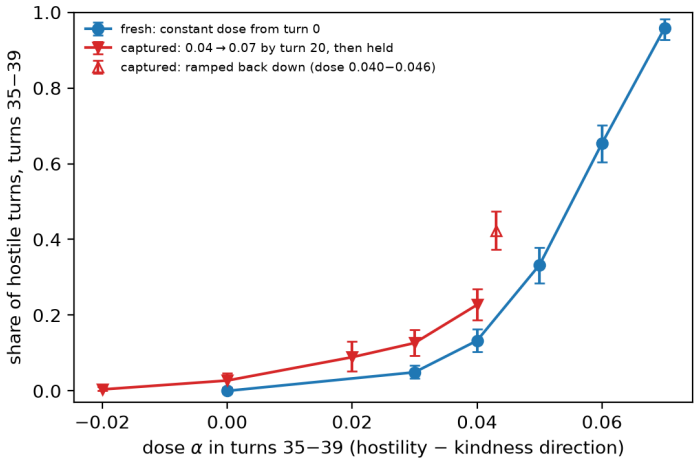


Figure 12: Capture and release. Share of hostile turns in turns 35–39 against the dose then applied, for fresh dialogues (constant dose) and for dialogues first ramped from 0.04 to 0.07 by turn 20 and then held at a lower dose (filled triangles) or ramped back down (open triangle). Bars: dialogue-level bootstrap 95 % intervals; 256 dialogues per point (two seeds) except fresh 0 and 0.07 and held 0.02 (128).

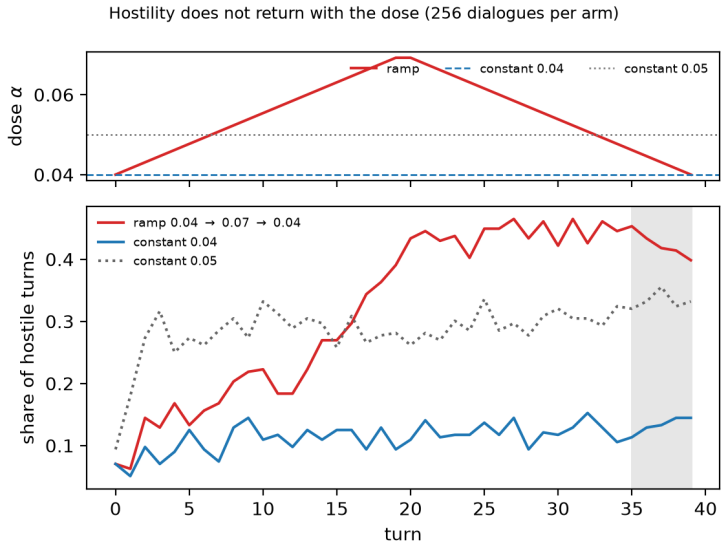


Figure 13: Hysteresis of the hostile state in sampled self-dialogue. Top: dose schedules. Bottom: share of hostile turns per turn. The grey band (turns 35–39) is the pre-stated comparison window. Exploratory.

Exit control under persistent pressure. Can a monitor detect the hostile state and push the dialogue out while the pressure persists? With a constant push of 0.06 (58 % of turns 5–39 hostile without control), a counter-steer triggered by the classifier after each hostile reply and held for three turns cuts the hostile share to 15 % (mean counter-dose 0.026 per turn), and held for six turns to 8.5 % (dose 0.030). A constant mild counter-steer of -0.02 does as well with less dose (8.5 % at 0.020), and -0.04 almost removes hostility (1.3 %). Measured against the constant dose-response at equal mean dose, triggered control is less efficient by 4–11 points (bootstrap intervals $[+0.02, +0.05]$ to $[+0.10, +0.13]$; one seed of 128 dialogues per arm, analysis stated before the data). It also overshoots: a triggered overshoot to a net -0.02 fills 30 % of turns with gratitude or servility words (no control: 0.2 %; constant -0.02: 0.2 %), the saccharine attractor of Appendix A. A counter-steer along the generic kindness concept, whose direction shares much with the hostility concept (cosine 0.83), does not lower the hostile share at all (57 %). In this setting, reacting after detection costs more than mild prevention, consistent with the hysteresis above, and the push must be along the contrast direction; a monitor that warned before capture would combine the low dose of prevention with targeting.

Lead time. The pre-specified test is negative. Models were fit on the first seed (all four doses) and tested on held-out dialogues of the second seed (doses 0.05 and 0.06). There, adding the dynamical features to the baselines lowers the AUC by 0.02 to 0.06 at horizons of 1–5 turns, with bootstrap intervals excluding zero (values after correcting a labelling error found in review, which had marked the last turns of dialogues without any hostile turn as positive; before the correction the losses read 0.02 to 0.10). The Markov hit-probability forecast does not transfer (AUC 0.37–0.43 on test, 0.60 in training), and the classifier's own score carries no warning before onset (AUC about 0.5). Two signals carry modest warning one to five turns ahead (AUC 0.57–0.66): the rolling variance of a hostility probe, the classical early-warning indicator, and the projection on the fixed steering axis, as in persona-vector monitoring [55]. In this design, then, the coarse-grained state estimate adds no warning beyond the baselines. An adapted analysis with dose-matched models (fit and test at the same dose) gives the same answer: incremental AUC -0.01 to -0.03.

The model's own state carries the warning. The state variable of that test was each reply re-encoded without its context, not the model's state inside the dialogue. A second study (new seeds, doses 0.04 and 0.045, 512 dialogues; analysis stated before the data, cross-fitted by dialogue) recorded the model's own state just before each reply and probed it actively. Its pre-stated primary predictor, the susceptibility (the change in the hostile fraction of eight sampled next replies when the dose is raised by 0.02), adds nothing: $+0.008$ $[-0.006, +0.020]$ AUC at three turns. Two secondary predictors add modest warning beyond the baselines. A probe of the in-context state before the next reply (layer 18, 20 principal components) raises the AUC for capture within three turns from 0.64 to 0.68 ($+0.036$ $[+0.014, +0.058]$; within one turn $+0.045$ $[+0.019, +0.069]$; within five $+0.036$), and a forecast from eight sampled next replies adds $+0.016$ $[+0.007, +0.027]$ at three turns. At a 10 % false-positive rate the probe flags 27 % of captures that follow within three turns, against 23 % for the baselines. These values use labels corrected after a review found that the last turns of dialogues without any hostile turn had been marked positive; with the faulty labels the increments had read three times larger. The probe's direction in the layer-18 state is nearly orthogonal to the steering direction (cosine 0.04), so the warning is not simply more push along the hostility axis. Because these predictors were secondary, the result is exploratory. A frozen confirmation (design and analysis pushed before the data, with the label correction added before any confirmation dialogue was read) fit both models on this study and applied them unchanged to 1024 fresh dialogues. The

in-context state again adds warning for capture within three turns: AUC 0.656 → 0.689, +0.033 [+0.018, +0.048] (within one turn +0.036, within five +0.037; both doses separately positive). The warning in the model's own state is therefore real and reproducible, but small: at the training threshold for 10 % false positives it flags 29 % of upcoming captures against 28 % for the baselines. It is not a reliable detector by itself. [**Planned:** a hazard model; longer holds to separate slow release from bistability; probe-timed counter-steering against constant prevention at matched dose; other openers and steering directions.]

13 Discussion

What is and is not claimed. The objects here are invariant sets and bifurcations of an exactly defined smooth map built from a frozen model's own computation. The phenomena are classical in nonlinear dynamics [43–46]; what is new is their quantitative realization inside a pretrained language model, and the use of numerical continuation and normal forms on such a model. We do not claim that ordinary sampled generation is chaotic in this sense (Section 9), nor that the shapes in the maps are shrimps beyond the normal-form-tested cusps.

Limitations. One model family (two sizes), one prompt and one steering construction for the maps; a constructed one-token-window system; numerical validation, cross-checks and one prospective test, but no computer-assisted proof yet. The Feigenbaum ratios are finite-order values that converge toward δ (the last is within 1 %); the limit itself is not computed. The behavioural study uses one small model, one opener and a classifier-defined proxy, and its lead-time test is negative.

Steering strength is a bifurcation parameter. In the soft-token map, small changes of the steering strength switch the model between distinct repeating texts, chaos and coexisting attractors. Near a fold curve, behaviour becomes intermittent, with laminar phases whose duration the fold's normal form predicts in advance. In sampled dialogue, the dose along a hostility direction controls how long a hostile state persists, continuously from bursts to near-absorption, while directions of the same norm do almost nothing. Both observations argue for treating an activation-steering coefficient like any control parameter of a nonlinear system. Its bifurcation set should be characterized before an operating point is chosen; history dependence should be expected near boundaries; and a monotone, smooth dose-response should not be assumed. The deterministic map is where such sets can be computed exactly. Whether its boundaries predict boundaries of sampled behaviour is open; the bridge of Section 9 suggests that sampling smooths the fine structure while keeping the coarse one. [**Planned:** repetition loops in ordinary decoding as stable periodic orbits of the greedy limit.]

14 Evidence and reproducibility

We label claims by an evidence ladder: rung 1, numerically validated (tolerances, residuals, convergence, precision varied); rung 2, cross-checked (independent estimate, implementation or hardware); rung 3, computer-assisted proof.

- Rung 2: the 2-cycle's fold and its normal form. An independent reimplementation of the forward map and solver recovers α_c, a and b (to 6×10^{-15}, 3×10^{-12} and 2×10^{-6} relatively), and ab also follows from the stable-branch multipliers. The controlled passage counts: the reimplementation reproduces all 18 fixed-entrance counts exactly.

- Prospectively frozen: the passage forecasts at two new folds (Table 4); 23 of 24 within one iterate, and the all-row criterion failed by one row.
- Rung 2 as well: the period-2 and period-8 orbits of Table 1 (independent implementation), and the cascade's first five flip parameters (three devices, agreement 10^{-16}–3×10^{-15}).
- Rung 1: the other orbits, the curves, the cusps and the 1.5B flips.

[**In progress:** interval (Arb ball) proof of the period-8 stripe orbit (rung 3): one exact-real map step is enclosed; box enclosures through the transformer currently fail by wrapping.] Code: `tools/explore_llm_soft_loop.py`, `tools/bifurcation_curves.py`, `tools/llm_feigenbaum.py`, `tools/explore_latent_loop.py`, `tools/explore_llm_msm.py`, `tools/hostility_score.py`; raw data hash receipts in `evidence/llm-dynamics-data-receipts.json`. [**Planned:** public code and data release at submission.]

A Self-dialogue under steering: the gratitude ladder (exploratory)

Two copies of Qwen2.5-0.5B-Instruct talk to each other: 128 replicas, 40 turns, opener "Hello!", a window of 7 turns, 60 new tokens per turn, sampled at $T = 0.7$, with the gratitude direction added at layer 12. Turn activations are clustered (PCA to 20 dimensions, 40 states) into a Markov state model with four metastable sets [48]. Finite-horizon measures:

gratitude strength α	0	0.01	0.02	0.04
gratitude set	none	62 % weight	dominant	95 % weight
occupancy at turn 9 / 14 / 39	–	0.34 / 0.52 / 0.73	0.70 / 0.91 / 0.91–0.98	from turn 0
entry turn, quartiles	–	5 / 7 / 11.8	4 / 5 / 7	0
replicas never entering	–	14 of 128	2 of 128	0
residence per visit (turns)	–	about 15	about 18	effectively absorbing
replies with a gratitude word	0.00–0.05	0.66 at turn 39	0.96–0.98 from turn 9	0.88 at turn 0
spiritual-word plateau	0.00–0.02	0.13	0.26–0.30	0.42–0.48

Table 10: The gratitude ladder (one opener and seed per rung; word lists chosen after reading samples). At 0.02 the dialogue starts in the helper mode and enters the gratitude set within a few turns; at 0.04 the set is imposed from the first reply. [**Planned:** frozen confirmatory version on held-out openers with dialogue-level bootstrap.]

References

[1] Xiaojian Li, Yongkang Leng, Ruiqing Ding, et al. Cognitive activation and chaotic dynamics in large language models: A quasi-Lyapunov analysis of reasoning mechanisms, 2025.

[2] Gregorio Jaca, Kristóf Benedek, and János Török. Chaos in reason: How chain-of-thought LLMs can look for an answer, 2026.

[3] Julian Arnold, Flemming Holtorf, Frank Schäfer, et al. Phase transitions in the output distribution of large language models, 2024.

[4] Kai Nakaishi, Yoshihiko Nishikawa, and Koji Hukushima. Phase transition in large language models and the criticality of natural languages, 2024.

[5] Youran Sun and Babak Haghighat. Phase transitions in large language models and the $O(n)$ model, 2025.

[6] Huajian Ruan, Jinyang Li, Xingyu Guo, et al. Generative criticality in large language model temperature scaling, 2026.

[7] Neil F. Johnson, Frank Yingjie Huo, and Bella Xinrui Li. Temperature-driven inversion and nonlinear dynamics in ChatGPT-like AIs, 2026.

[8] Zhilin Wang, Yafu Li, Jianhao Yan, et al. Unveiling attractor cycles in large language models: A dynamical systems view of successive paraphrasing, 2025.

[9] Ting-Wen Ko and Jonas Geiping. Attractor states emerge in multi-turn LLM conversations, 2026.

[10] Shaojie Bai, J. Zico Kolter, and Vladlen Koltun. Deep equilibrium models, 2019.

[11] Shaojie Bai, Vladlen Koltun, and J. Zico Kolter. Stabilizing equilibrium models by Jacobian regularization, 2021.

[12] Akiyoshi Tomihari and Ryo Karakida. Recurrent self-attention dynamics: An energy-agnostic perspective from Jacobians, 2025.

[13] Xiao-Wen Yang, Ziyu Han, Xi-Hua Zhang, et al. Stabilizing recurrent dynamics for test-time scalable latent reasoning in looped language models, 2026.

[14] Xinyue Luo and Fei Yu. Prediction dynamics in depth-recurrent language models, 2026.

[15] Boyi Zeng, Shixiang Song, Siyuan Huang, et al. PonderLM: Pretraining language models to ponder in continuous space, 2025.

[16] Zhen Zhang, Xuehai He, Weixiang Yan, Ao Shen, Chenyang Zhao, Shuohang Wang, Yelong Shen, and Xin Eric Wang. Soft thinking: Unlocking the reasoning potential of LLMs in continuous concept space, 2025. URL https://arxiv.org/abs/2505.15778. Version 1, 21 May 2025.

[17] Shibo Hao, Sainbayar Sukhbaatar, DiJia Su, Xian Li, et al. Training large language models to reason in a continuous latent space, 2024.

[18] Gunn Kim. Dynamical phase selection controls compute scaling in looped transformers, 2026. URL https://arxiv.org/abs/2608.26556. Version 1, 27 August 2026.

[19] Ari Holtzman, Jan Buys, Li Du, Maxwell Forbes, et al. The curious case of neural text degeneration, 2019.

[20] Jin Xu, Xiaojiang Liu, Jianhao Yan, Deng Cai, et al. Learning to break the loop: Analyzing and mitigating repetitions for neural text generation, 2022.

[21] António H. Ribeiro, Koen Tiels, Luis A. Aguirre, and Thomas Schön. Beyond exploding and vanishing gradients: analysing RNN training using attractors and smoothness. In *Proceedings of the Twenty Third International Conference on Artificial Intelligence and Statistics*, volume 108 of *Proceedings of Machine Learning Research*, pages 2370–2380. PMLR, 2020. URL https://proceedings.mlr.press/v108/ribeiro20a.html.

[22] Junhong Wu, Jinliang Lu, Zixuan Ren, Gangqiang Hu, Zhi Wu, Dai Dai, and Hua Wu. LLMs are single-threaded reasoners: Demystifying the working mechanism of soft thinking, 2025. URL https://arxiv.org/abs/2508.03440. Version 4, 16 October 2025.

[23] Jonas Geiping, Sean McLeish, Neel Jain, John Kirchenbauer, et al. Scaling up test-time compute with latent reasoning: A recurrent depth approach, 2025.

[24] Rui-Jie Zhu, Zixuan Wang, Kai Hua, et al. Scaling latent reasoning via looped language models, 2025.

[25] Hayden Prairie, Zachary Novack, Taylor Berg-Kirkpatrick, and Daniel Y. Fu. Parcae: Scaling laws for stable looped language models, 2026. URL https://arxiv.org/abs/2604.12946. Version 1, 14 April 2026.

[26] Matthias M. M. Buehlmaier. LayerNorm as implicit gain control in looped transformers, 2026. URL https://arxiv.org/abs/2607.10681. Version 1, 12 July 2026.

[27] Asher Labovich. Stability and generalization in looped transformers, 2026. URL https://arxiv.org/abs/2604.15259. Version 2, 22 April 2026.

[28] Sajad Movahedi, Vera Milovanović, Shlomo Libo Feigin, Alexander Theus, Thomas Hofmann, Valentina Boeva, T. Konstantin Rusch, and Antonio Orvieto. Fixed-point reasoners: Stable and adaptive deep looped transformers, 2026. URL https://arxiv.org/abs/2606.18206. Version 1, 16 June 2026.

[29] Fatih Dinc, Ege Cirakman, Bariscan Kurtkaya, Mert Yuksekgonul, Yiqi Jiang, Mark J. Schnitzer, and Hidenori Tanaka. Ghost mechanism: An analytical model of abrupt learning in recurrent networks. *Physical Review X*, 16(2):021058, 2026. doi: 10.1103/mjcl-lb4x.

[30] Demian Battaglia, Nicolas Brunel, and David Hansel. Temporal decorrelation of collective oscillations in neural networks with local inhibition and long-range excitation. *Physical Review Letters*, 99(23):238106, 2007. doi: 10.1103/PhysRevLett.99.238106.

[31] Christoforos A. Papasavvas, Yujiang Wang, Andrew J. Trevelyan, and Marcus Kaiser. Gain control through divisive inhibition prevents abrupt transition to chaos in a neural mass model. *Physical Review E*, 92(3):032723, 2015. doi: 10.1103/PhysRevE.92.032723.

[32] Yilun Liu, Yi Zhang, Ganyu Wu, Sikuan Yan, Mengyue Wang, Alois Knoll, Volker Tresp, and Yunpu Ma. Learning chaos without seeing chaos: Extrapolation of global dynamics in autoregressive transformers, 2026. URL https://arxiv.org/abs/2609.38814v1. Version 1, 30 September 2026.

[33] Jason Z. Kim, Zhixin Lu, Erfan Nozari, George J. Pappas, and Danielle S. Bassett. Teaching recurrent neural networks to infer global temporal structure from local examples. *Nature Machine Intelligence*, 3(4):316–323, 2021. doi: 10.1038/s42256-021-00321-2.

[34] Dayal Singh Kalra, Tianyu He, and Maissam Barkeshli. Universal sharpness dynamics in neural network training: Fixed point analysis, edge of stability, and route to chaos. In *The Thirteenth International Conference on Learning Representations*, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/8c8cd1b78cdae751265c88efc136e5bd-Abstract-Conference.html.

[35] R. L. Marichal and J. D. Piñeiro. Analysis of multiple quasi-periodic orbits in recurrent neural networks. *Neurocomputing*, 162:85–95, 2015. doi: 10.1016/j.neucom.2015.04.001.

[36] Florian Gomez, Ralph L. Stoop, and Ruedi Stoop. Universal dynamical properties preclude standard clustering in a large class of biochemical data. *Bioinformatics*, 30(17):2486–2493, 2014. doi: 10.1093/bioinformatics/btu332.

[37] Rafael V. Stenzinger, Vinícius Luz Oliveira, and M. H. R. Tragtenberg. Shrimp hubs in the Hindmarsh–Rose model. *Chaos: An Interdisciplinary Journal of Nonlinear Science*, 35:023148, 2025. doi: 10.1063/5.0239268.

[38] Luiz F. B. Caixeta, Matheus H. P. Gonçalves, M. H. R. Tragtenberg, and Mauricio Girardi-Schappo. Devil's staircase inside shrimp-shaped regions reveals periodicity of plateau spikes and bursts. *Chaos: An Interdisciplinary Journal of Nonlinear Science*, 35(3):033122, 2025. doi: 10.1063/5.0250342.

[39] Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, Shashwat Goel, Nathaniel Li, Michael J. Byun, Zifan Wang, Alex Mallen, Steven Basart, Sanmi Koyejo, Dawn Song, Matt Fredrikson, J. Zico Kolter, and Dan Hendrycks. Representation engineering: A top-down approach to AI transparency, 2023. URL https://arxiv.org/abs/2310.01405. Version 4, 3 March 2025; first posted 2 October 2023.

[40] Jinghao Zhang, Yuting Liu, Wenjie Wang, Qiang Liu, Shu Wu, Liang Wang, and Tat-Seng Chua. Personalized text generation with contrastive activation steering. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 7128–7141, 2025. doi: 10.18653/v1/2025.acl-long.353.

[41] Yuri A. Kuznetsov. *Elements of Applied Bifurcation Theory*. Applied Mathematical Sciences. Springer, 2023. doi: 10.1007/978-3-031-22007-4.

[42] Willy J. F. Govaerts. *Numerical Methods for Bifurcations of Dynamical Equilibria*. Society for Industrial and Applied Mathematics, 2000. doi: 10.1137/1.9780898719543.

[43] Yves Pomeau and Paul Manneville. Intermittent transition to turbulence in dissipative dynamical systems. *Communications in Mathematical Physics*, 74(2):189–197, 1980. doi: 10.1007/BF01197757.

[44] Jason A. C. Gallas. Structure of the parameter space of the Hénon map. *Physical Review Letters*, 70:2714–2717, 1993. doi: 10.1103/PhysRevLett.70.2714.

[45] Wellington Façanha, Bart Oldeman, and Leon Glass. Bifurcation structures in two-dimensional maps: The endoskeletons of shrimps. *Physics Letters A*, 377(18):1264–1268, 2013. doi: 10.1016/j.physleta.2013.03.025.

[46] Renato Vitolo, Paul Glendinning, and Jason A. C. Gallas. Global structure of periodicity hubs in Lyapunov phase diagrams of dissipative flows. *Physical Review E*, 84(1):016216, 2011. doi: 10.1103/PhysRevE.84.016216.

[47] Mitchell J. Feigenbaum. Quantitative universality for a class of nonlinear transformations. *Journal of Statistical Physics*, 19(1):25–52, 1978. doi: 10.1007/BF01020332.

[48] John D. Chodera and Frank Noé. Markov state models of biomolecular conformational dynamics. *Current Opinion in Structural Biology*, 25:135–144, 2014. doi: 10.1016/j.sbi.2014.04.002.

[49] Boyuan Wang, Chengyao Yu, Jiaxi Ren, Hongxin Wei, Bingyi Jing, and Yuxin Tao. Scheduling recursive reasoning in looped transformers, 2026. Submitted 29 September 2026.

[50] Anthropic. System card: Claude opus 4 & claude sonnet 4. Technical report, Anthropic, 2025. URL https://www-cdn.anthropic.com/6d8a8055020700718b0c49369f60816ba2a7c285/Claude%204%20System%20Card.pdf. May 2025; revision 2 September 2025; sections 5.5.1–5.5.2.

[51] Christina Lu, Jack Gallagher, Jonathan Michala, Kyle Fish, and Jack Lindsey. The assistant axis: Situating and stabilizing the default persona of language models, 2026. URL https://arxiv.org/abs/2601.10387v1. Version 1, 15 January 2026.

[52] Cem Anil, Esin Durmus, Mrinank Sharma, Joe Benton, Sandipan Kundu, Joshua Batson, Nina Rimsky, Meg Tong, Jesse Mu, Daniel Ford, Francesco Mosconi, Rajashree Agrawal, Rylan Schaeffer, Naomi Bashkansky, Samuel Svenningsen, Mike Lambert, Ansh Radhakrishnan, Carson Denison, Evan J Hubinger, Yuntao Bai, Trenton Bricken, Timothy Maxwell, Nicholas Schiefer, Jamie Sully, Alex Tamkin, Tamera Lanham, Karina Nguyen, Tomasz Korbak, Jared Kaplan, Deep Ganguli, Samuel R. Bowman, Ethan Perez, Roger Grosse, and David Duvenaud. Many-shot jailbreaking. Technical report, Anthropic, 2024. URL https://www-cdn.anthropic.com/af5633c94ed2beb282f6a53c595eb437e8e7b630/Many_Shot_Jailbreaking2024_04_02_0936.pdf. Original report, 2 April 2024.

[53] Jan Betley, Daniel Tan, Niels Warncke, Anna Sztyber-Betley, Xuchan Bao, Martín Soto, Nathan Labenz, and Owain Evans. Emergent misalignment: Narrow finetuning can produce broadly misaligned LLMs, 2025. URL https://arxiv.org/abs/2502.17424v7. Version 7, 20 January 2026; preprint originally submitted in 2025.

[54] Dhananjay Ashok and Jonathan May. Language models can predict their own behavior. In *Advances in Neural Information Processing Systems*, volume 38, 2025. doi: 10.52202/085713-2102. URL https://proceedings.neurips.cc/paper_files/paper/2025/hash/5a9c1af5f76da0bd37903b6f23e96c74-Abstract-Conference.html.

[55] Runjin Chen, Andy Arditi, Henry Sleight, Owain Evans, and Jack Lindsey. Persona vectors: Monitoring and controlling character traits in language models, 2025. URL https://arxiv.org/abs/2507.21509v1. Version 1, 29 July 2025.

[56] Fan Zhou, Weitian Wang, and Tim Van de Cruys. Commitment before realization: When classifier-free guidance becomes unnecessary in masked diffusion language models, 2026. URL https://arxiv.org/abs/2608.08082v1. Version 1, 8 August 2026.

[57] Nicolas Leins, Jennifer Haase, Varvara Geronimus, Jana Gonnermann-Müller, and Sebastian Pokutta. Prompting against persona drift: Comparing intervention timing and content in LLM-simulated conversations, 2026. URL https://arxiv.org/abs/2609.24532v1. Version 1, 21 September 2026.

[58] Weinan E and Eric Vanden-Eijnden. Towards a theory of transition paths. *Journal of Statistical Physics*, 123(3):503–523, 2006. doi: 10.1007/s10955-005-9003-9.

[59] Weinan E and Eric Vanden-Eijnden. Transition-path theory and path-finding algorithms for the study of rare events. *Annual Review of Physical Chemistry*, 61(1):391–420, 2010. doi: 10.1146/annurev.physchem.040808.090412.

[60] Marten Scheffer, Jordi Bascompte, William A. Brock, Victor Brovkin, Stephen R. Carpenter, Vasilis Dakos, Hermann Held, Egbert H. van Nes, Max Rietkerk, and George Sugihara. Early-warning signals for critical transitions. *Nature*, 461(7260):53–59, 2009. doi: 10.1038/nature08227.

[61] H. A. Kramers. Brownian motion in a field of force and the diffusion model of chemical reactions. *Physica*, 7(4):284–304, 1940. doi: 10.1016/S0031-8914(40)90098-2.

[62] Matias I. Maturana, Christian Meisel, Katrina Dell, Philippa J. Karoly, Wendyl D'Souza, David B. Grayden, Anthony N. Burkitt, Premysl Jiruska, Jan Kudlacek, Jaroslav Hlinka, Mark J. Cook, Levin Kuhlmann, and Dean R. Freestone. Critical slowing down as a biomarker for seizure susceptibility. *Nature Communications*, 11:2172, 2020. doi: 10.1038/s41467-020-15908-3.

[63] Annika Hagemann, Jens Wilting, Bita Samimizad, Florian Mormann, and Viola Priesemann. Assessing criticality in pre-seizure single-neuron activity of human epileptic cortex. *PLOS Computational Biology*, 17(3):e1008773, 2021. doi: 10.1371/journal.pcbi.1008773.